

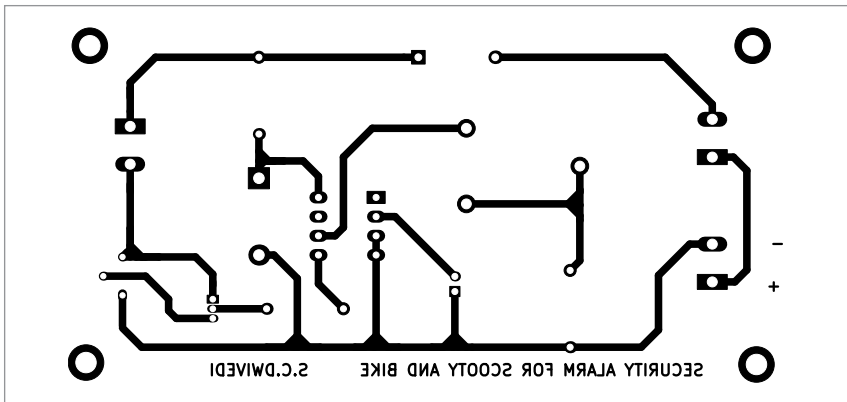


Fig. 3: Actual-size, single-side PCB design for the circuit

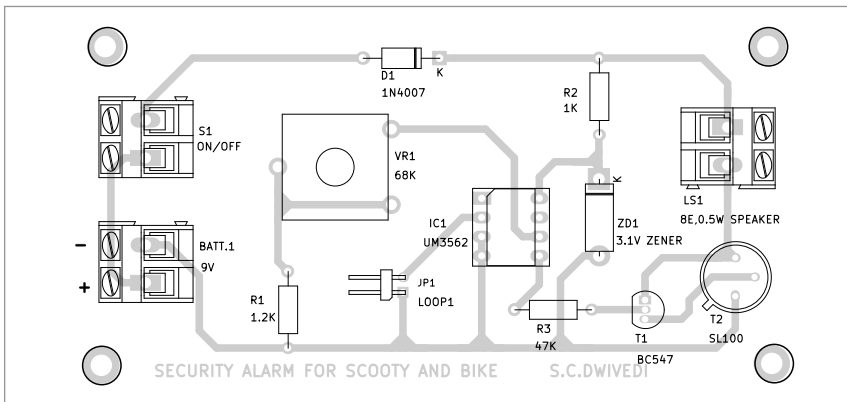


Fig. 4: Component layout of the PCB



Fig. 5: Author's prototype in a box

PARTS LIST

Semiconductors:

- IC1 - UM3562 machine-gun sound generator
- T1 - BC547 NPN transistor
- T2 - SL100 NPN transistor
- D1 - 1N4007 rectifier diode
- ZD1 - 3.1V Zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

- R1 - 1.2-kilo-ohm
- R2 - 1-kilo-ohm
- R3 - 47-kilo-ohm
- VR1 - 68-kilo-ohm preset

Miscellaneous:

- S1 - On/off switch
- LS1 - 8-ohm, 0.5-watt speaker
- BATT.1 - 9V battery
- Thin flexible wire

After connecting the circuit to bike's wheel, switch on S1 to enable the circuit. This readies the circuit to protect the bike. When someone tries to drive away the bike and breaks the wire, loudspeaker LS1 sounds an alert to let you know someone is trying to steal your bike.

Construction and testing

An actual-size, single-side PCB for the bike alarm is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Connect the flexible wire tightly to the bike's wheel such that it would break if someone turns the bike's handle to drive it away.

After connecting battery to the circuit, switch on S1 and hide the assembled unit somewhere in the bike (say, under the spare tyre). Fix the thin flexible wire in such a way that it will definitely break when someone turns the vehicle's handle to move it. The author's prototype wired on a general-purpose PCB and fitted in a plastic box is shown in Fig. 5. **EFY**

BONUS. You can watch the VIDEO of the tutorial of this DIY project at: <https://www.electronicsforu.com/videos-slideshows/live-diy-security-alarm-scooty-bike>

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